

ROAD GUARDIAN AI - PROJECT IDEA SUBMISSION

Real-Time Road Health & Traffic Digital Twin System | Smart City & AI Infrastructure

PROJECT TITLE: Road Guardian AI: Real-Time Road Health & Traffic Digital Twin

Submission Date: August 09, 2026 | Domain: AI, Smart Cities, Urban Transportation

1. EXECUTIVE SUMMARY

Road Guardian AI is an end-to-end autonomous road infrastructure surveillance and predictive traffic intelligence platform. The system transforms road maintenance from reactive manual inspection into proactive predictive intelligence. By combining real-time YOLO computer vision with contextual risk evaluation (speed, weather, traffic density, and school/hospital proximity), the system maps citywide road health onto a 3D Digital Twin map and simulates how closing damaged roads for repair will impact surrounding traffic flow before maintenance begins.

2. PROBLEM STATEMENT

1. Reactive Manual Surveys: Municipalities rely on slow citizen complaints or periodic surveys to find severe potholes.
2. Context-Blind Risk Assessment: Traditional tools treat a pothole on a quiet lane the same as one on a 90 km/h expressway. In reality, speed, rain (turning potholes into invisible puddles), and traffic volume drastically alter the real-world risk.
3. Repair Gridlock: Closing an arterial road for emergency repair without predictive planning causes severe traffic spillover, congestion bottlenecks, and emergency vehicle delays.

3. PROPOSED 4-LAYER ARCHITECTURE

Layer 1 - Perception (AI Computer Vision):

- Uses YOLOv8 & OpenCV to detect potholes, cracks, and surface damage in real time with EXIF GPS logging.

Layer 2 - Context Engine (Road Risk Score 0-100):

- Multi-factor algorithm evaluating Damage (35%), Speed (20%), Traffic (15%), Road Type (10%), Weather (10%), Proximity (10%).

Layer 3 - Digital Twin City Road Network Map:

- 3D spatial WebGL graph mapping road segments into Green (Healthy), Yellow (Degraded), Orange (High Risk), Red (Critical).

Layer 4 - Predictive Traffic Intelligence & Rerouting Simulator:

- Simulates road closures and calculates exact traffic shifts (e.g. Closing Road A increases traffic on Road B by +46.8%).

4. KEY INNOVATIONS & TECHNICAL STACK

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| - Core Language: Python 3.10+ | - AI Perception: YOLOv8 PyTorch & OpenCV |
| - Web Framework: Streamlit Dashboard | - 3D Mapping: PyDeck (Deck.gl WebGL) |
| - Risk Engine: Custom Weighted Math Model | - Traffic Rerouting: Graph Flow Math Engine |
| - Reporting: FPDF2 PDF Generator | - Municipal Portal: API Payload & Ticket Logger |

5. BUSINESS IMPACT & MUNICIPAL VALUE

- Cost Efficiency: Saves up to 40% in municipal road maintenance costs by prioritizing critical risk roads first.
- Accident Prevention: Reduces speed/weather hazard collisions through automated risk alerts.
- Gridlock Elimination: Helps traffic police adjust green-light signal timing and set up detour signs before repair closures start.

Official Idea Submission Document - Road Guardian AI